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D20A - 33

Experiment: 09

Aim: To implement a Private Ethereum Blockchain using Geth.

Theory:

Ethereum is an open-source, decentralized blockchain platform that enables developers to build and deploy smart contracts and decentralized applications (DApps). Unlike public blockchains such as the Ethereum mainnet, a Private Ethereum Blockchain is restricted to specific participants, making it ideal for development, testing, and enterprise applications where confidentiality, control, and performance are essential.

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through JSON-RPC, command-line, or JavaScript consoles.

Setting up a **Private Ethereum Network** involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

Steps Involved:

1. **Choosing a Network ID:**
2. Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
3. **Choosing a Consensus Algorithm:**
4. Ethereum supports different consensus mechanisms such as **Proof of Work (PoW)** and **Proof of Authority (PoA)**. In private networks, **Clique (PoA)** is often used for faster block confirmation.
5. **Creating a Genesis Block:**
6. The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
7. **Initializing the Geth Database:**
8. The `geth init` command initializes the blockchain using the **genesis.json** file, preparing the data directory for node operation.
9. **Setting up Networking:**
10. Nodes communicate using **enodes** (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
11. **Running the Member Nodes:**

12. Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
13. **Running a Signer (in Clique):**
14. In the **Clique consensus**, designated **signers** validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

Implementation:

1. First install all the necessary dependencies like curl, git, build-essential.
sudo apt install wget curl git build-essential -y

```
root@admission-5: ~        root@admission-5: ~
root@admission-5:~# sudo apt install wget curl git build-essential -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.8).
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 73 not upgraded.
root@admission-5:~#
```

2. Install geth using wget
cd /usr/local/bin && sudo wget
https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e5 01b3b0.tar.gz

```
root@admission-5:/usr/local/bin# sudo wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1
.12.0-e501b3b0.tar.gz
--2026-04-07 05:38:24--  https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)|20.60.40.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 26856444 (26M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.12.0-e501b3b0.tar.gz'

geth-linux-amd64-1.12.0-e5 100%[======================================>]  25.61M  2.28MB/s    in 9.3s

2026-04-07 05:38:32 (2.77 MB/s) - 'geth-linux-amd64-1.12.0-e501b3b0.tar.gz' saved [26856444/26856444]

root@admission-5:/usr/local/bin#
```

3. Extract the tar file and move the geth into /usr/local/bin/geth path & check the geth version
sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz sudo mv
geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth geth version

```
root@admission-5:/usr/local/bin# sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz
sudo mv geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth
geth version
geth-linux-amd64-1.12.0-e501b3b0/
geth-linux-amd64-1.12.0-e501b3b0/COPYING
geth-linux-amd64-1.12.0-e501b3b0/geth
Geth
Version: 1.12.0-stable
Git Commit: e501b3b05db8e169f67dc78b7b59bc352b3c638d
Git Commit Date: 20230525
Architecture: amd64
Go Version: go1.20.3
Operating System: linux
GOPATH=
GOROOT=
root@admission-5:/usr/local/bin#
```

4. Create a folder named private_ethereum_setup and create another 2 directories naming node1 and node2 individually inside private_ethereum_setup directory.
mkdir private_ethereum_setup cd private_ethereum_setup && mkdir node1 node2

```
root@admission-5:/usr/local/bin# mkdir private_ethereum_setup
root@admission-5:/usr/local/bin# cd private_ethereum_setup && mkdir node1 node2
root@admission-5:/usr/local/bin/private_ethereum_setup# ls
node1  node2
root@admission-5:/usr/local/bin/private_ethereum_setup#
```

5. Create account for node1
geth --datadir "node1" account new

```
root@admission-5:/usr/local/bin/private_ethereum_setup# geth --datadir "node1" account new
INFO [04-07|05:42:29.002] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-07|05:42:29.003] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0xB39388323dCD82eeb28F73B3448424c008FE7150
Path of the secret key file: node1/keystore/UTC--2026-04-07T05-42-42.824513266Z--b39388323dcd82eeb28f73b3448424c008fe7150

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

root@admission-5:/usr/local/bin/private_ethereum_setup#
```

6. Do the same for node2
geth --datadir "node2" account new

```
root@admission-5:/usr/local/bin/private_ethereum_setup# geth --datadir "node2" account new
INFO [04-07|05:43:28.087] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-07|05:43:28.087] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x126F82349979c6b3d2a52eE032B7ea11880D2dC9
Path of the secret key file: node2/keystore/UTC--2026-04-07T05-43-35.386933441Z--126f82349979c6b3d2a52ee032b7ea11880d2dc9

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

root@admission-5:/usr/local/bin/private_ethereum_setup#
```

7. Create a file called genesis.json in the same directory and copy paste these contents

```
{
 "config": {
 "chainId": 12345,
 "homesteadBlock": 0,
 "eip150Block": 0,
 "eip155Block": 0,
 "eip158Block": 0,
 "byzantiumBlock": 0,
 "constantinopleBlock": 0,
 "petersburgBlock": 0,
 "istanbulBlock": 0,
 "berlinBlock": 0,
 "londonBlock": 0,
 "clique": {
 "period": 5,
 "epoch": 30000
 }
 },
 "difficulty": "1",
 "gasLimit": "8000000",
 "extradata":
"0x00000000000000000000000000000000000000000000000000000000
00000000c99cf64ef49425bf0e814c9d51364281a61747d200000000000
000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000
00000",
 "alloc": {
 "0xB39388323dCD82eeb28F73B3448424c008FE7150": {
 "balance": "100000000000000000000"
```

},
"0x126F82349979c6b3d2a52eE032B7ea11880D2dC9": {
"balance": "100000000000000000000"
}
}
}

8. Cat genesis.json

```
root@admission-5:/usr/local/bin/private_ethereum_setup# cat genesis.json
{
 "config": {
 "chainId": 12345,
 "homesteadBlock": 0,
 "eip150Block": 0,
 "eip155Block": 0,
 "eip158Block": 0,
 "byzantiumBlock": 0,
 "constantinopleBlock": 0,
 "petersburgBlock": 0,
 "istanbulBlock": 0,
 "berlinBlock": 0,
 "londonBlock": 0,
 "clique": {
 "period": 5,
 "epoch": 30000
 }
 },
 "difficulty": "1",
 "gasLimit": "8000000",
 "extradata":
"0x00000000000000000000000000000000000000000000000000000000
00000000c99cf64ef49425bf0e814c9d51364281a61747d200000000000
00000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000
00000",
 "alloc": {
 "0xB39388323dCD82eeb28F73B3448424c008FE7150": {
 "balance": "100000000000000000000"
 },
 "0x126F82349979c6b3d2a52eE032B7ea11880D2dC9": {
 "balance": "100000000000000000000"
```

9. Initializing geth for node1 using the genesis.json file
geth init --datadir node1 genesis.json

```
}root@admission-5:/usr/local/bin/private_ethereum_setup# geth init --datadir node1 genesis.json
INFO [04-07|05:58:25.537] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-07|05:58:25.538] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-07|05:58:25.543] Set global gas cap                       cap=50,000,000
INFO [04-07|05:58:25.544] Initializing the KZG library             backend=gokzg
INFO [04-07|05:58:25.594] Defaulting to pebble as the backing database
INFO [04-07|05:58:25.595] Allocated cache and file handles         database=/usr/local/bin/private_ethereum_setup/node1/geth/chaindat
a cache=16.00MiB handles=16
INFO [04-07|05:58:25.618] Opened ancient database                  database=/usr/local/bin/private_ethereum_setup/node1/geth/chaindat
a/ancient/chain readonly=false
INFO [04-07|05:58:25.619] Writing custom genesis block
INFO [04-07|05:58:25.619] Persisted trie from memory database      nodes=3 size=411.00B time="9.642µs" gcnodes=0 gcsize=0.00B gctime=
0s livenodes=0 livesize=0.00B
INFO [04-07|05:58:25.623] Successfully wrote genesis state         database=chaindata hash=8c5dd7..24caed
INFO [04-07|05:58:25.623] Defaulting to pebble as the backing database
INFO [04-07|05:58:25.623] Allocated cache and file handles         database=/usr/local/bin/private_ethereum_setup/node1/geth/lightcha
indata cache=16.00MiB handles=16
INFO [04-07|05:58:25.640] Opened ancient database                  database=/usr/local/bin/private_ethereum_setup/node1/geth/lightcha
indata/ancient/chain readonly=false
INFO [04-07|05:58:25.641] Writing custom genesis block
INFO [04-07|05:58:25.641] Persisted trie from memory database      nodes=3 size=411.00B time="10.121µs" gcnodes=0 gcsize=0.00B gctime
=0s livenodes=0 livesize=0.00B
INFO [04-07|05:58:25.644] Successfully wrote genesis state         database=lightchaindata hash=8c5dd7..24caed
root@admission-5:/usr/local/bin/private_ethereum_setup#
```

10. Do the same for node2

geth init --datadir node2 genesis.json

```
root@admission-5:/usr/local/bin/private_ethereum_setup# geth init --datadir node2 genesis.json
INFO [04-07|05:59:40.725] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-07|05:59:40.726] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-07|05:59:40.729] Set global gas cap                       cap=50,000,000
INFO [04-07|05:59:40.729] Initializing the KZG library             backend=gokzg
INFO [04-07|05:59:40.760] Defaulting to pebble as the backing database
INFO [04-07|05:59:40.760] Allocated cache and file handles         database=/usr/local/bin/private_ethereum_setup/node2/geth/chaindat
a cache=16.00MiB handles=16
INFO [04-07|05:59:40.780] Opened ancient database                  database=/usr/local/bin/private_ethereum_setup/node2/geth/chaindat
a/ancient/chain readonly=false
INFO [04-07|05:59:40.780] Writing custom genesis block
INFO [04-07|05:59:40.780] Persisted trie from memory database      nodes=3 size=411.00B time="10.186µs" gcnodes=0 gcsize=0.00B gctime
=0s livenodes=0 livesize=0.00B
INFO [04-07|05:59:40.784] Successfully wrote genesis state         database=chaindata hash=8c5dd7..24caed
INFO [04-07|05:59:40.784] Defaulting to pebble as the backing database
INFO [04-07|05:59:40.784] Allocated cache and file handles         database=/usr/local/bin/private_ethereum_setup/node2/geth/lightcha
indata cache=16.00MiB handles=16
INFO [04-07|05:59:40.805] Opened ancient database                  database=/usr/local/bin/private_ethereum_setup/node2/geth/lightcha
indata/ancient/chain readonly=false
INFO [04-07|05:59:40.805] Writing custom genesis block
INFO [04-07|05:59:40.805] Persisted trie from memory database      nodes=3 size=411.00B time="10.09µs"  gcnodes=0 gcsize=0.00B gctime
=0s livenodes=0 livesize=0.00B
INFO [04-07|05:59:40.808] Successfully wrote genesis state         database=lightchaindata hash=8c5dd7..24caed
root@admission-5:/usr/local/bin/private_ethereum_setup#
```

11. Now we will install bootnode for configuring both the nodes.

sudo add-apt-repository -y ppa:ethereum/ethereum

sudo apt-get update

sudo apt-get install bootnode

```
root@admission-5:/usr/local/bin/private_ethereum_setup# sudo add-apt-repository -y ppa:ethereum/ethereum
sudo apt-get update
sudo apt-get install bootnode
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu/
Suites: noble
Components: main
'
More info: https://launchpad.net/~ethereum/+archive/ubuntu/ethereum
Adding repository.
Hit:1 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble InRelease [18.1 kB]
Get:7 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main amd64 Packages [2576 B]
Get:8 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main Translation-en [772 B]
Fetched 21.4 kB in 1s (18.3 kB/s)
Reading package lists... Done
Hit:1 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  bootnode
0 upgraded, 1 newly installed, 0 to remove and 73 not upgraded.
```

12. Create a key for the bootnode and save it to boot.key in the folder
bootnode -genkey boot.key

```
root@admission-5:/usr/local/bin/private_ethereum_setup# bootnode -genkey boot.key
root@admission-5:/usr/local/bin/private_ethereum_setup# ls
boot.key  genesis.json  node1  node2
root@admission-5:/usr/local/bin/private_ethereum_setup#
```

13. Copy this command and copy the enode://…. This will be required later
bootnode -nodekey boot.key -verbosity 9 -addr :30305

```
root@admission-5:/usr/local/bin/private_ethereum_setup# bootnode -nodekey boot.key -verbosity 9 -addr :30305
enode://7ff7a90d47bc73762802a4f924fdcb403af4bff08ded3c3b0b5c2efeb9f76553859f6edf9abdf20619c04c9408fc7ed606e097a26d4519d50f5afcc063e02
120@127.0.0.1:0?discport=30305
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [04-07|06:03:30.800] New local node record                    seq=1,775,541,810,798 id=f572303c2cd66d79 ip="invalid IP" udp=0 tc
p=0
```

14. Now on another terminal we will start mining node1
geth --datadir node1 \
--port 30306 \
--bootnodes
"enode://7ff7a90d47bc73762802a4f924fdcb403af4bff08ded3c3b0b5c2efeb9f76553859f6edf9ab
df20619c04c9408fc7ed606e097a26d4519d50f5afcc063e02120@127.0.0.1:30305" \
--networkid 123454321 \

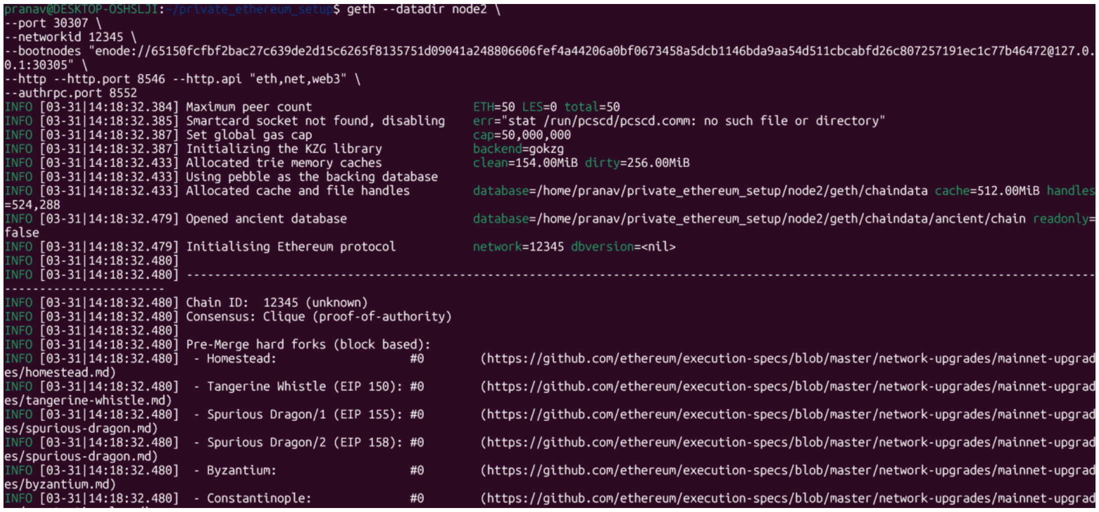
--unlock 0xB39388323dCD82eeb28F73B3448424c008FE7150 \
--password node1/password.txt \
--authrpc.port 8551 \
--miner.etherbase 0xB39388323dCD82eeb28F73B3448424c008FE7150 \
--mine

```
root@admission-5:/usr/local/bin/private_ethereum_setup# rm -rf /usr/local/bin/private_ethereum_setup/node1/geth
root@admission-5:/usr/local/bin/private_ethereum_setup# geth --datadir node1 --port 30306 --bootnodes "enode://7ff7a90d47bc73762802a4
f924fdcb403af4bff08ded3c3b0b5c2efeb9f76553859f6edf9abdf20619c04c9408fc7ed606e097a26d4519d50f5afcc063e02120@127.0.0.1:30305" --network
id 123454321 --unlock 0xB39388323dCD82eeb28F73B3448424c008FE7150 --password node1/password.txt --authrpc.port 8551 --miner.etherbase
0xB39388323dCD82eeb28F73B3448424c008FE7150 --mine
INFO [04-07|06:38:47.164] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-07|06:38:47.166] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-07|06:38:47.168] Set global gas cap                       cap=50,000,000
INFO [04-07|06:38:47.169] Initializing the KZG library             backend=gokzg
INFO [04-07|06:38:47.199] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-07|06:38:47.199] Defaulting to pebble as the backing database
INFO [04-07|06:38:47.199] Allocated cache and file handles         database=/usr/local/bin/private_ethereum_setup/node1/geth/chaindat
a cache=512.00MiB handles=524,288
INFO [04-07|06:38:47.225] Opened ancient database                  database=/usr/local/bin/private_ethereum_setup/node1/geth/chaindat
a/ancient/chain readonly=false
INFO [04-07|06:38:47.225] Initialising Ethereum protocol           network=123,454,321 dbversion=<nil>
INFO [04-07|06:38:47.226] Writing default main-net genesis block
INFO [04-07|06:38:47.461] Persisted trie from memory database      nodes=12356 size=1.79MiB time=48.92926ms gcnodes=0 gcsize=0.00B gc
time=0s livenodes=0 livesize=0.00B
INFO [04-07|06:38:47.476]
INFO [04-07|06:38:47.476] ----------------------------------------------------------------------------------------------------------
---------------------------------------------
INFO [04-07|06:38:47.476] Chain ID:  1 (mainnet)
INFO [04-07|06:38:47.476] Consensus: Beacon (proof-of-stake), merged from Ethash (proof-of-work)
INFO [04-07|06:38:47.476]
INFO [04-07|06:38:47.476] Pre-Merge hard forks (block based):
INFO [04-07|06:38:47.476]  - Homestead:                   #1150000  (https://github.com/ethereum/execution-specs/blob/master/network-
upgrades/mainnet-upgrades/homestead.md)
INFO [04-07|06:38:47.476]  - DAO Fork:                    #1920000  (https://github.com/ethereum/execution-specs/blob/master/network-
upgrades/mainnet-upgrades/dao-fork.md)
INFO [04-07|06:38:47.476]  - Tangerine Whistle (EIP 150): #2463000  (https://github.com/ethereum/execution-specs/blob/master/network-
upgrades/mainnet-upgrades/tangerine-whistle.md)
```

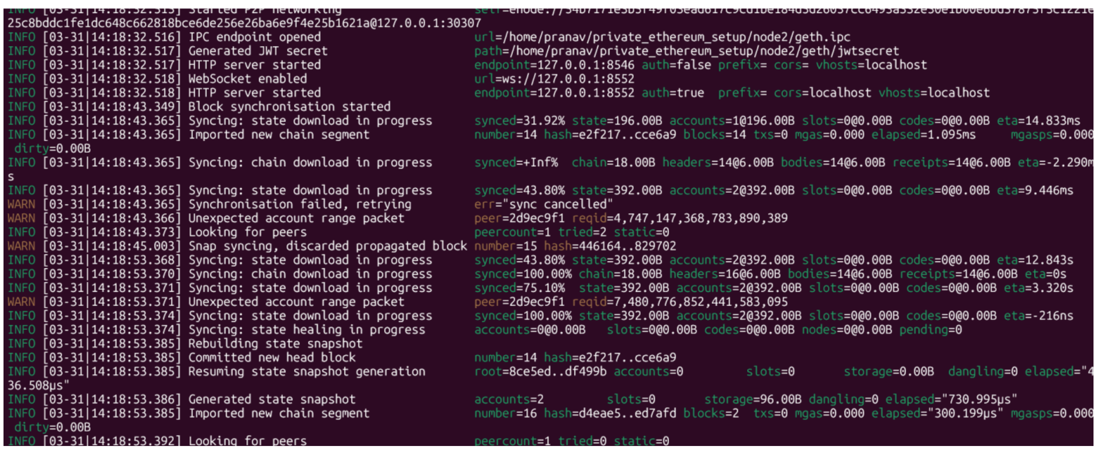
```
root@admission-5: ~    root@admission-5: /usr/local/l    root@admission-5: /usr/local,
WARN [04-07|06:53:41.791] Unclean shutdown detected                booted=2026-04-07T06:38:47+0000 age=14m54s
WARN [04-07|06:53:41.791] Engine API enabled                       protocol=eth
INFO [04-07|06:53:41.791] Starting peer-to-peer node               instance=Geth/v1.12.0-stable-e501b3b0/linux-amd64/go1.20.3
INFO [04-07|06:53:41.799] Generated state snapshot                 accounts=8893 slots=0 storage=409.64KiB dangling=0 elapsed=9.332ms
INFO [04-07|06:53:41.806] New local node record                    seq=1,775,543,927,501 id=5ffab40659cdeaba ip=127.0.0.1 udp=30306 t
cp=30306
INFO [04-07|06:53:41.806] Started P2P networking                   self=enode://45d557b46f2495e3c7bdfd4c01c5e3025b83fbbfe5ef89badb57a
00a7d87a8d4c6ccab873cdb8e40fb59e0d5b9ed3063d1a2e7b1c0042de2863090ad672c3b45@127.0.0.1:30306
INFO [04-07|06:53:41.807] IPC endpoint opened                      url=/usr/local/bin/private_ethereum_setup/node1/geth.ipc
INFO [04-07|06:53:41.807] Loaded JWT secret file                   path=/usr/local/bin/private_ethereum_setup/node1/geth/jwtsecret cr
c32=0x21e9937f
INFO [04-07|06:53:41.808] WebSocket enabled                        url=ws://127.0.0.1:8551
INFO [04-07|06:53:41.808] HTTP server started                      endpoint=127.0.0.1:8551 auth=true prefix= cors=localhost vhosts=lo
calhost
INFO [04-07|06:53:42.426] Unlocked account                         address=0xB39388323dCD82eeb28F73B3448424c008FE7150
INFO [04-07|06:53:42.426] Transaction pool price threshold updated price=0
INFO [04-07|06:53:42.426] Transaction pool price threshold updated price=1,000,000,000
INFO [04-07|06:53:42.427] Commit new sealing work                  number=1 sealhash=0dc167..0b853c uncles=0 txs=0 gas=0 fees=0 elaps
ed="787.008µs"
panic: ethash (pow) sealing not supported any more

goroutine 46 [running]:
github.com/ethereum/go-ethereum/consensus/ethash.(*Ethash).Seal(0xc0002f6520?, {0x41aea6?, 0xc00044c2c0?}, 0xc00044c3c8?, 0xc00044c2c
0?, 0xf4a6bcad6d815a31?)
        github.com/ethereum/go-ethereum/consensus/ethash/ethash.go:84 +0x27
github.com/ethereum/go-ethereum/consensus/beacon.(*Beacon).Seal(0xc000496a80, {0x2066f20, 0xc000581800}, 0xc0002c6640, 0x9daddf69ca14
7c60?, 0x9e5c859464211001?)
        github.com/ethereum/go-ethereum/consensus/beacon/consensus.go:387 +0x9e
github.com/ethereum/go-ethereum/miner.(*worker).taskLoop(0xc0001ce280)
        github.com/ethereum/go-ethereum/miner/worker.go:706 +0x2fc
created by github.com/ethereum/go-ethereum/miner.newWorker
        github.com/ethereum/go-ethereum/miner/worker.go:329 +0xa8d
root@admission-5:/usr/local/bin/private_ethereum_setup#
```

30°C Sunny Search ENG US 12:24 07-04-2026

15. Now do same for node2



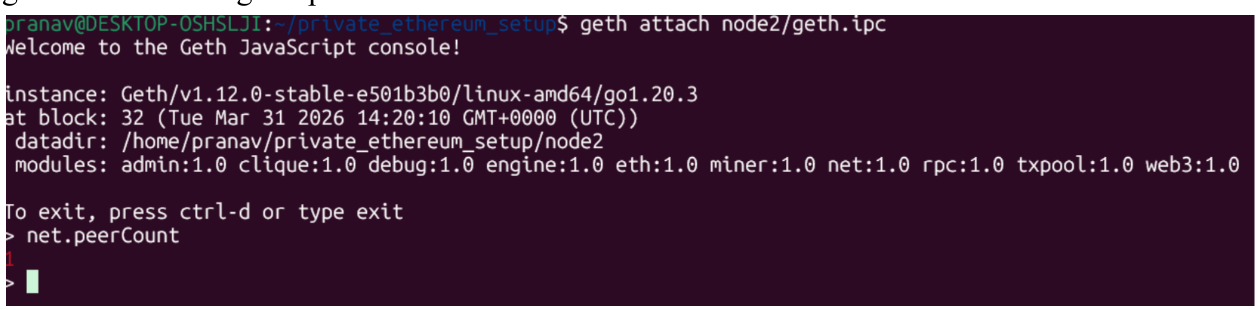
```
pranav@DESKTOP-OSHSLJI:~/private_ethereum_setup$ geth --datadir node2 \
--port 30307 \
--networkid 12345 \
--bootnodes "enode://65150fcfbf2bac27c639de2d15c6265f8135751d09041a248806606fef4a44206a0bf0673458a5dcb1146bda9aa54d511cbcabfd26c807257191ec1c77b46472@127.0.0.1:30305" \
--http --http.port 8546 --http.api "eth,net,web3" \
--authrpc.port 8552
INFO [03-31|14:18:32.384] Maximum peer count                       ETH=50 LES=0 total=50
INFO [03-31|14:18:32.385] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-31|14:18:32.387] Set global gas cap                       cap=50,000,000
INFO [03-31|14:18:32.387] Initializing the KZG library             backend=gokzg
INFO [03-31|14:18:32.433] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [03-31|14:18:32.433] Using pebble as the backing database
INFO [03-31|14:18:32.433] Allocated cache and file handles         database=/home/pranav/private_ethereum_setup/node2/geth/chaindata cache=512.00MiB handles=524,288
INFO [03-31|14:18:32.479] Opened ancient database                  database=/home/pranav/private_ethereum_setup/node2/geth/chaindata/ancient/chain readonly=false
INFO [03-31|14:18:32.479] Initialising Ethereum protocol           network=12345 dbversion=<nil>
INFO [03-31|14:18:32.480]
INFO [03-31|14:18:32.480] ---------------------------------------------------------------------------------------------------------------------------------------------------
INFO [03-31|14:18:32.480] Chain ID:  12345 (unknown)
INFO [03-31|14:18:32.480] Consensus: Clique (proof-of-authority)
INFO [03-31|14:18:32.480]
INFO [03-31|14:18:32.480] Pre-Merge hard forks (block based):
INFO [03-31|14:18:32.480]  - Homestead:                   #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [03-31|14:18:32.480]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistle.md)
INFO [03-31|14:18:32.480]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [03-31|14:18:32.480]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.md)
INFO [03-31|14:18:32.480]  - Byzantium:                   #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/byzantium.md)
INFO [03-31|14:18:32.480]  - Constantinople:              #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrad
```



```
25c8bddc1fe1dc648c662818bce6de256e26ba6e9f4e25b1621a@127.0.0.1:30307
INFO [03-31|14:18:32.516] IPC endpoint opened                      url=/home/pranav/private_ethereum_setup/node2/geth.ipc
INFO [03-31|14:18:32.517] Generated JWT secret                     path=/home/pranav/private_ethereum_setup/node2/geth/jwtsecret
INFO [03-31|14:18:32.517] HTTP server started                      endpoint=127.0.0.1:8546 auth=false prefix= cors= vhosts=localhost
INFO [03-31|14:18:32.518] WebSocket enabled                        url=ws://127.0.0.1:8552
INFO [03-31|14:18:32.518] HTTP server started                      endpoint=127.0.0.1:8552 auth=true  prefix= cors=localhost vhosts=localhost
INFO [03-31|14:18:43.349] Block synchronisation started
INFO [03-31|14:18:43.365] Syncing: state download in progress      synced=31.92% state=196.00B accounts=1@196.00B slots=0@0.00B codes=0@0.00B eta=14.833ms
INFO [03-31|14:18:43.365] Imported new chain segment               number=14 hash=e2f217..cce6a9 blocks=14 txs=0 mgas=0.000 elapsed=1.095ms     mgasps=0.000 dirty=0.00B
INFO [03-31|14:18:43.365] Syncing: chain download in progress      synced=+Inf%  chain=18.00B headers=14@6.00B bodies=14@6.00B receipts=14@6.00B eta=-2.290ms
INFO [03-31|14:18:43.365] Syncing: state download in progress      synced=43.80% state=392.00B accounts=2@392.00B slots=0@0.00B codes=0@0.00B eta=9.446ms
WARN [03-31|14:18:43.365] Synchronisation failed, retrying         err="sync cancelled"
WARN [03-31|14:18:43.366] Unexpected account range packet          peer=2d9ec9f1 reqid=4,747,147,368,783,890,389
INFO [03-31|14:18:43.373] Looking for peers                        peercount=1 tried=2 static=0
WARN [03-31|14:18:45.003] Snap syncing, discarded propagated block number=15 hash=446164..829702
INFO [03-31|14:18:53.368] Syncing: state download in progress      synced=43.80% state=392.00B accounts=2@392.00B slots=0@0.00B codes=0@0.00B eta=12.843s
INFO [03-31|14:18:53.370] Syncing: chain download in progress      synced=100.00% chain=18.00B headers=16@6.00B bodies=14@6.00B receipts=14@6.00B eta=0s
INFO [03-31|14:18:53.371] Syncing: state download in progress      synced=75.10%  state=392.00B accounts=2@392.00B slots=0@0.00B codes=0@0.00B eta=3.320s
WARN [03-31|14:18:53.371] Unexpected account range packet          peer=2d9ec9f1 reqid=7,480,776,852,441,583,095
INFO [03-31|14:18:53.374] Syncing: state download in progress      synced=100.00% state=392.00B accounts=2@392.00B slots=0@0.00B codes=0@0.00B eta=-216ns
INFO [03-31|14:18:53.374] Syncing: state healing in progress       accounts=0@0.00B   slots=0@0.00B codes=0@0.00B nodes=0@0.00B pending=0
INFO [03-31|14:18:53.385] Rebuilding state snapshot
INFO [03-31|14:18:53.385] Committed new head block                 number=14 hash=e2f217..cce6a9
INFO [03-31|14:18:53.385] Resuming state snapshot generation       root=8ce5ed..df499b accounts=0         slots=0       storage=0.00B  dangling=0 elapsed="436.508µs"
INFO [03-31|14:18:53.386] Generated state snapshot                 accounts=2         slots=0       storage=96.00B dangling=0 elapsed="730.995µs"
INFO [03-31|14:18:53.385] Imported new chain segment               number=16 hash=d4eae5..ed7afd blocks=2  txs=0 mgas=0.000 elapsed="300.199µs" mgasps=0.000 dirty=0.00B
INFO [03-31|14:18:53.392] Looking for peers                        peercount=1 tried=0 static=0
```

16. Now we will interact with the blockchain using javascript console

geth attach node1/geth.ipc



```
pranav@DESKTOP-OSHSLJI:~/private_ethereum_setup$ geth attach node2/geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.12.0-stable-e501b3b0/linux-amd64/go1.20.3
at block: 32 (Tue Mar 31 2026 14:20:10 GMT+0000 (UTC))
 datadir: /home/pranav/private_ethereum_setup/node2
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> net.peerCount
1
>
```

17. Interacting with the blockchain

```
> eth.getBalance(eth.accounts[0])
1e+21
> web3.fromWei(eth.getBalance("0x563a5a670260417b163ad1696fcbfa6849906d06"), "ether")
0
> eth.getBlock(eth.blockNumber)
{
  baseFeePerGas: 10300,
  difficulty: 2,
  extraData: "0xd883010c00846765746888676f312e32302e33856c696e757800000000004331cba2f966b7937d8d70eb13d05086a2a85d6e26959dfb0e771f0987e27f32
0eb990949208332f2cdc5e7aecbf80ad7392c1d7541ec88859cec7cfb4c14aca01",
  gasLimit: 8700408,
  gasUsed: 0,
  hash: "0x0fdff9b1594109661ff4afdf42b06c14107de1d8c4ecb279a42e41fa1a240a5b",
  logsBloom: "0x00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000",
  miner: "0x0000000000000000000000000000000000000000",
  mixHash: "0x0000000000000000000000000000000000000000000000000000000000000000",
  nonce: "0x0000000000000000",
  number: 86,
  parentHash: "0xdcfd357a39f38bacc798691a64e8a9943c1b66799c7aa299ef5b3173aa13c08f",
  receiptsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  sha3Uncles: "0x1dcc4de8dec75d7aab85b567b6ccd41ad312451b948a7413f0a142fd40d49347",
  size: 609,
  stateRoot: "0x8ce5edc7603ca4790ca7556cd0e3da011d6297bda73f93599ebc78fb35df499b",
  timestamp: 1774967080,
  totalDifficulty: 173,
  transactions: [],
  transactionsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  uncles: []
}
```

18. Sending transactions between accounts

```
> eth.sendTransaction({
......     from: "0xC99cf64ef49425Bf0e814c9d51364281a61747d2",
......     to: "0x19f1609855626d5551a589497acb92966c69c778",
......     value: web3.toWei(1, "ether")
...... })
"0x9f724e5b7bd747a7815130974af859a25ed67984d1869c9372a4707f451edc6f"
```

19. Analyzing transactions

```
> eth.getTransaction("0x9f724e5b7bd747a7815130974af859a25ed67984d1869c9372a4707f451edc6f")
{
  accessList: [],
  blockHash: "0x8dbf4c55125132f781e4eba86dbcb71508732210046a78eddcc7ce655226140c",
  blockNumber: 328,
  chainId: "0x3039",
  from: "0xc99cf64ef49425bf0e814c9d51364281a61747d2",
  gas: 21000,
  gasPrice: 1000000007,
  hash: "0x9f724e5b7bd747a7815130974af859a25ed67984d1869c9372a4707f451edc6f",
  input: "0x",
  maxFeePerGas: 1000000014,
  maxPriorityFeePerGas: 1000000000,
  nonce: 0,
  r: "0xe8f59df6f1949c3822df1c43a8958ef1e64450607413e7fd4cffeb3641b066c4",
  s: "0x1f100bc05d1038738501ea097087193d905f810423f4261960a16229d1ccec4c",
  to: "0x19f1609855626d5551a589497acb92966c69c778",
  transactionIndex: 0,
  type: "0x2",
  v: "0x1",
  value: 1000000000000000000
}
```

```
> eth.getBlock(355)
{
  baseFeePerGas: 7,
  difficulty: 2,
  extraData: "0xd883010c00846765746888676f312e32302e33856c696e75780000000000000054e881ad4052a191aaff4b0e0424dc223f194b9c05561
5bfb86960bc387b199878645b27b2f6cd8e758771fd09e0d1c1de0c63a7ebf214219ae29612c057212600",
  gasLimit: 11312359,
  gasUsed: 0,
  hash: "0x687ab31e5c2c7d347d56020734532d38028c0bc6907f622033620759711fbc08",
  logsBloom: "0x0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000",
  miner: "0x0000000000000000000000000000000000000000",
  mixHash: "0x0000000000000000000000000000000000000000000000000000000000000000",
  nonce: "0x0000000000000000",
  number: 355,
  parentHash: "0xe0da19a2c7dbd84c2356a7d5430f06d7b8759a91647a84fd813277b5af2dd778",
  receiptsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  sha3Uncles: "0x1dcc4de8dec75d7aab85b567b6ccd41ad312451b948a7413f0a142fd40d49347",
  size: 609,
  stateRoot: "0x4d859554d617ddf6e59474da15f2438b0485ef0ce98b7dc17405739a73258b05",
  timestamp: 1774976575,
  totalDifficulty: 711,
  transactions: [],
  transactionsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  uncles: []
}
```

Conclusion

Private blockchain was created using geth, the nodes in the blockchain were configured and we interacted with the blockchain by making a transaction of 10 ethers.